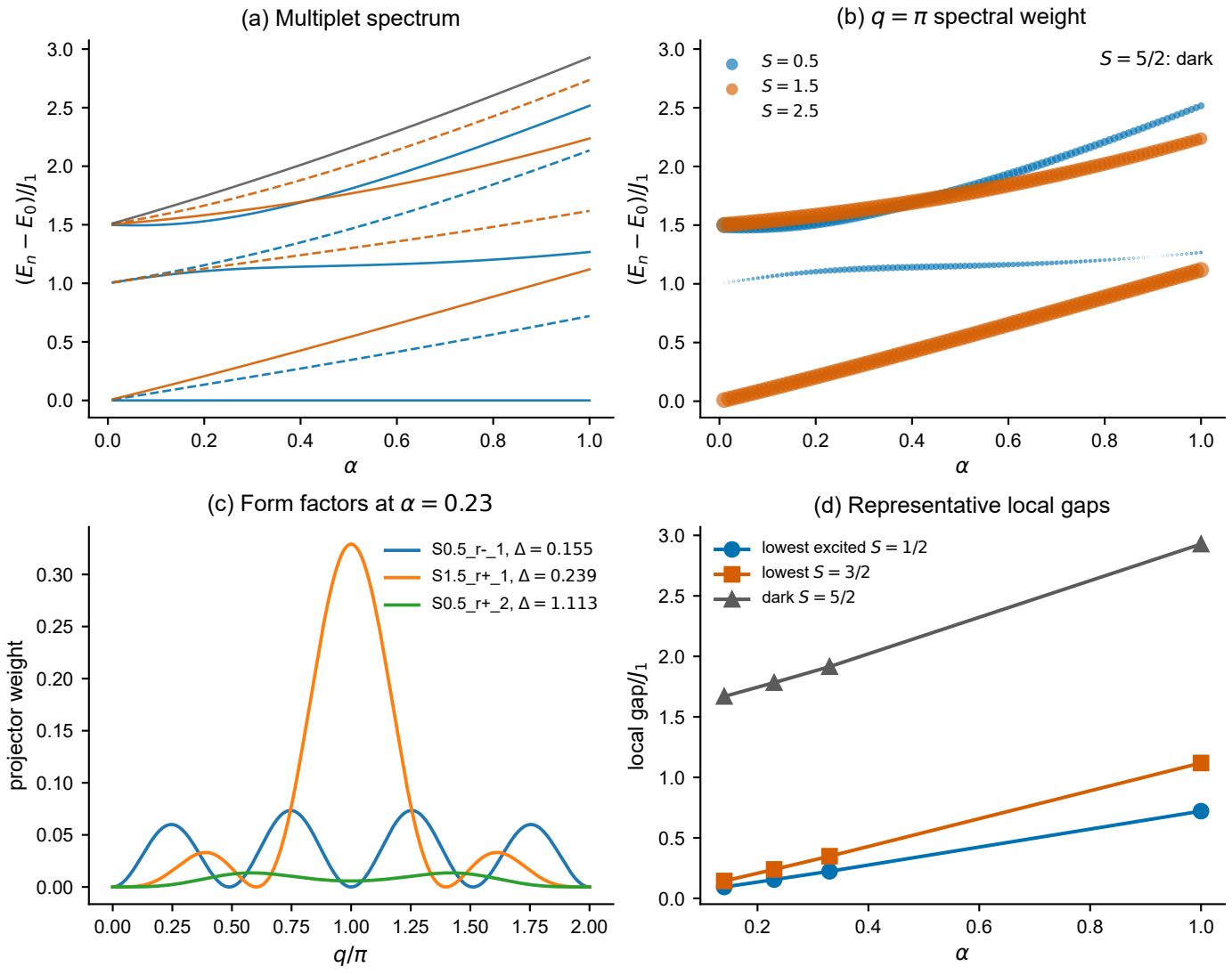

(a) Multiplet spectrum
$(E_n - E_0)/J_1$
α
(b) $q = \pi$ spectral weight
$S = 0.5$
$S = 1.5$
$S = 2.5$
$S = 5/2$: dark
$(E_n - E_0)/J_1$
α
(c) Form factors at $\alpha = 0.23$
S0.5_r-_1, $\Delta = 0.155$
S1.5_r+_1, $\Delta = 0.239$
S0.5_r+_2, $\Delta = 1.113$
projector weight
q/π
(d) Representative local gaps
lowest excited $S = 1/2$
lowest $S = 3/2$
dark $S = 5/2$
local gap/J_1
α